

CY202 - Model Question Papers

C. V. RAMAN GLOBAL UNIVERSITY, ODISHA

Department of Chemistry

CY202: BIOLOGY AND ENVIRONMENTAL SCIENCE

Ten Model Question Paper Sets - aligned to the official CY202 syllabus

Format (as per the university question papers): Part A - 15 questions x 2 marks, all compulsory. Part B - Questions 2 to 6; Q2, Q3 and Q4 carry three bits (answer any two) and Q5 and Q6 carry two bits (answer the following). Total: 100 marks, 3 hours.

How the ten sets are arranged: Sets 1 to 5 keep the module order fixed (Q2 = Module 1, Q3 = Module 2, Q4 = Module 3, Q5 = Module 4, Q6 = Module 5). Sets 6 to 8 keep each module once but shuffle the order. Sets 9 and 10 are fully randomised - the questions are drawn at random from all modules, while the Part A and Part B structure is unchanged. Part A always carries three questions from each module, in module order.

Coverage (per the syllabus): Unit 1 - cells, tissues, cell organelles, comparative plant/animal/bacterial cell, carbohydrates, proteins, fats, nucleic acids, enzymes, chromosomes and genes. Unit 2 - gene expression, DNA replication, transcription and translation, mutation, enzymes as biocatalysts, human health and diseases, antibiotics, immunity and vaccination, HIV and AIDS, drugs and alcohol abuse, medicinal plants. Unit 3 - ecosystem structure and functions, energy flow, food chain and food web, biogeochemical cycles, environmental gradients, tolerance levels, E.I.A. Unit 4 - air pollutants, greenhouse gases, industrial emission control, air-pollution policy, noise sources and standards. Unit 5 - water quality and treatment, COD and BOD, solid waste and ISWM, hazardous waste, e-waste. Questions compiled from the five unit question banks and the previous-year question papers (Summer 2025, Regular 2025, End-Semester 2025 and 2026).

C. V. RAMAN GLOBAL UNIVERSITY, ODISHA
MODEL QUESTION PAPER - SET 1
 END SEMESTER EXAMINATION - BIOLOGY AND ENVIRONMENTAL SCIENCE
 Part B in module order (Q2 = Module 1 ... Q6 = Module 5)

Subject Code: CY202	Subject: BIOLOGY AND ENVIRONMENTAL SCIENCE
Branch: B.Tech - ALL BRANCHES	Time: 3 Hours Full Marks: 100

Answer all Questions from Part - A and Part - B.

Part-A (Answer all questions)				
Q	Part-A (Answer all questions)	CO	Marks	BL
(a)	Define a cell and name the scientist credited with its discovery.	1	2	1
(b)	List any four characteristics of living organisms.	1	2	2
(c)	State two structural differences between prokaryotic and eukaryotic cells.	1	2	3
(d)	What are enzymes? Write down any two applications of enzymes.	2	2	3
(e)	What are the different ways of enzyme inhibition?	2	2	1
(f)	Define fermentation and write the equation for alcoholic fermentation.	2	2	1
(g)	What is the difference between environment and ecosystem?	3	2	3
(h)	Define ecosystem.	3	2	3
(i)	Mention the types of components of the ecosystem with examples.	3	2	3
(j)	What do you mean by air pollution?	4	2	2
(k)	Define particulate matter.	4	2	3
(l)	Mention the division of particulate matter with dimension.	4	2	3
(m)	What are the three types of water quality parameters? Mention with examples.	5	2	2
(n)	What do you mean by Total Dissolved Solids (TDS)? How can TDS be used to measure the ionic conductance of water?	5	2	3
(o)	What is the saturation concentration of Dissolved Oxygen (DO) at 20°C and 0°C?	5	2	2

Part-B				
Q2. Answer any two:				
(a)	Describe the fundamental characteristics of living organisms.	1	7	3
(b)	Elaborate on the different types of organic constituents present in living cells with examples.	1	7	2
(c)	Compare and contrast prokaryotic and eukaryotic cells.	1	7	2
Q3. Answer any two:				
(a)	What are enzymes? Discuss their structure, characteristics and significance as biocatalysts in biological processes.	2	7	2
(b)	Define fermentation. Discuss the different steps of fermentation.	2	7	3
(c)	Define bioenergy. What are its different forms? Write the production processes and challenges of bioenergy.	2	7	2
Q4. Answer any two:				
(a)	Discuss the biotic components of the ecosystem and their organisation.	3	7	4
(b)	Discuss in detail the functions of the ecosystem.	3	7	2
(c)	What are trophic levels? Discuss each trophic level with example.	3	7	4
Q5. Answer the following:				
(a)	Discuss the characteristics, primary source and harmful effects of the criterion air pollutants.	4	7	3
(b)	What is acid rain? Discuss, with reactions, the harmful effects of acid rain.	4	7	3
Q6. Answer the following:				
(a)	What are the three phases of waste water treatment? Discuss, with diagrams, the working principle of the trickling filter.	5	7	4
(b)	What are biological oxidation ponds? Discuss their working principle with a proper diagram.	5	7	3

Best of Luck

C. V. RAMAN GLOBAL UNIVERSITY, ODISHA
MODEL QUESTION PAPER - SET 2
 END SEMESTER EXAMINATION - BIOLOGY AND ENVIRONMENTAL SCIENCE
 Part B in module order (Q2 = Module 1 ... Q6 = Module 5)

Subject Code: CY202	Subject: BIOLOGY AND ENVIRONMENTAL SCIENCE
Branch: B.Tech - ALL BRANCHES	Time: 3 Hours Full Marks: 100

Answer all Questions from Part - A and Part - B.

Part-A (Answer all questions)

Q	Part-A (Answer all questions)	CO	Marks	BL
(a)	Why is mitochondria known as the powerhouse of the cell?	1	2	2
(b)	Write down any two important functions of mitochondria.	1	2	2
(c)	What are the 'suicide bags' of the cell, and why are they so called?	1	2	2
(d)	Name the diseases that appear due to the deficiency of Vitamin A and Vitamin C.	2	2	3
(e)	Name the disease that appears due to the deficiency of Calcium and Iodine.	2	2	1
(f)	Write down any two advantages of bioenergy.	2	2	2
(g)	What are the levels of organisation in an ecosystem? Mention the hierarchy.	3	2	1
(h)	What do you mean by environmental sustainability?	3	2	3
(i)	What are NPP and GPP? What is their relation?	3	2	3
(j)	Name the major six air pollutants.	4	2	2
(k)	What are trace pollutants?	4	2	2
(l)	What are the criteria for gaseous pollutants in urban regions?	4	2	2
(m)	Mention any two methods of determination of Dissolved Oxygen (DO) in water.	5	2	1
(n)	What is eutrophication?	5	2	3
(o)	What is activated sludge?	5	2	2

Part-B**Q2. Answer any two:**

(a)	Describe the functions of the cell membrane, cell wall and cytoplasm.	1	7	2
(b)	Describe the structure and functions of cell organelles such as the endoplasmic reticulum, Golgi apparatus, mitochondria and lysosomes.	1	7	3
(c)	What are the differences between plant cells and animal cells?	1	7	3

Q3. Answer any two:

(a)	What is photosynthesis? Discuss the different steps involved in photosynthesis.	2	7	4
(b)	What are biofertilizers? How are they applied? Write down two advantages and two challenges.	2	7	4
(c)	What are pesticides? Write down four advantages and four disadvantages of pesticides.	2	7	2

Q4. Answer any two:

(a)	How does energy flow in an ecosystem? Discuss elaborately.	3	7	4
(b)	Discuss the carbon cycle in detail.	3	7	4
(c)	Describe the processes involved in the nitrogen cycle.	3	7	2

Q5. Answer the following:

(a)	What is global warming? Discuss the impact of greenhouse gases on global warming.	4	7	4
(b)	What is ozone layer depletion? How does CFC affect ozone depletion?	4	7	4

Q6. Answer the following:

(a)	Describe the major components of a hazardous waste management plan.	5	7	4
(b)	What is e-waste? Cite their common sources and composition. Discuss the steps involved in e-waste recycling.	5	7	3

Best of Luck

C. V. RAMAN GLOBAL UNIVERSITY, ODISHA
MODEL QUESTION PAPER - SET 3
 END SEMESTER EXAMINATION - BIOLOGY AND ENVIRONMENTAL SCIENCE
 Part B in module order (Q2 = Module 1 ... Q6 = Module 5)

Subject Code: CY202	Subject: BIOLOGY AND ENVIRONMENTAL SCIENCE
Branch: B.Tech - ALL BRANCHES	Time: 3 Hours Full Marks: 100

Answer all Questions from Part - A and Part - B.

Part-A (Answer all questions)

Q	Part-A (Answer all questions)	CO	Marks	BL
(a)	Define ribosome and write its function.	1	2	2
(b)	Write down two differences between rough and smooth endoplasmic reticulum.	1	2	3
(c)	State the principal function of the Golgi apparatus.	1	2	2
(d)	Write the general equation of photosynthesis and state where it mainly occurs in a plant cell.	2	2	3
(e)	What are biofertilizers?	2	2	3
(f)	What are pesticides?	2	2	1
(g)	State the laws of thermodynamics followed in energy flow in an ecosystem.	3	2	3
(h)	What do you mean by nutrient recycling?	3	2	1
(i)	Which biogeochemical cycle does not directly involve air or the atmosphere?	3	2	1
(j)	What is photochemical smog, and how is it different from smog?	4	2	1
(k)	What are VOCs?	4	2	3
(l)	Give the full forms: NAAQS, EPA, VOC, CFC, CCS.	4	2	1
(m)	Discuss in brief the working principle of a rotating biological contactor.	5	2	3
(n)	Mention the goals of circular economy.	5	2	2
(o)	What are the principles of 3R?	5	2	2

Part-B**Q2. Answer any two:**

(a)	Write down the differences between animal cells and bacterial cells.	1	7	3
(b)	Define cell. Write down five characteristics of cells and draw a labelled diagram of a cell.	1	7	4
(c)	What are enzymes? Discuss their structure, characteristics and working mechanism in the body.	1	7	3

Q3. Answer any two:

(a)	What are vaccines? Discuss the working principle of any three broad categories of vaccines.	2	7	2
(b)	What are steroids? Discuss the functions and significance of steroids with examples.	2	7	4
(c)	Discuss the common Indian medicinal plants and their uses.	2	7	4

Q4. Answer any two:

(a)	Discuss Environmental Impact Assessment (EIA).	3	7	3
(b)	What is a food web? Discuss with a suitable diagram.	3	7	4
(c)	What are limiting factors in an environment? Explain a tolerance curve for a single species population existing on a single environmental gradient.	3	7	3

Q5. Answer the following:

(a)	Write a short note on flue gas desulfurization (FGD).	4	7	4
(b)	Write a short note on noise pollution.	4	7	4

Q6. Answer the following:

(a)	What is Electromagnetic Radiation (EMR)? Mention its two types. Write the impacts of EMR exposure on human health and explain in brief the mitigation techniques.	5	7	4
(b)	The results from a BOD test, diluted by 100, are given in the table below. Calculate BOD_3 , BOD_5 and BOD_7 .	5	7	3

Table for Q6(b):

Time (days)	Diluted sample DO (mg/L)
0	7.95
1	3.75
3	2.75
5	1.80
7	0.75

Best of Luck

C. V. RAMAN GLOBAL UNIVERSITY, ODISHA
MODEL QUESTION PAPER - SET 4
 END SEMESTER EXAMINATION - BIOLOGY AND ENVIRONMENTAL SCIENCE
 Part B in module order (Q2 = Module 1 ... Q6 = Module 5)

Subject Code: CY202	Subject: BIOLOGY AND ENVIRONMENTAL SCIENCE
Branch: B.Tech - ALL BRANCHES	Time: 3 Hours Full Marks: 100

Answer all Questions from Part - A and Part - B.

Part-A (Answer all questions)

Q	Part-A (Answer all questions)	CO	Marks	BL
(a)	State any two differences between plant cells and animal cells.	1	2	2
(b)	Write down any two differences between animal cells and bacterial cells.	1	2	2
(c)	Mention two examples of disaccharides with their constituent monosaccharides.	1	2	1
(d)	Give two examples each of insecticides, herbicides and rodenticides.	2	2	1
(e)	What do you mean by biological pest control?	2	2	1
(f)	What are steroids? Name two examples.	2	2	3
(g)	What is a food chain? Explain in brief with a schematic diagram.	3	2	2
(h)	What do you mean by nitrogen fixation?	3	2	3
(i)	What do you mean by nitrification in the nitrogen cycle?	3	2	2
(j)	What happened during the Bhopal gas tragedy?	4	2	2
(k)	What are the optimum sizes of particulate matter that can be filtered by cyclones, scrubbers and electrostatic precipitators?	4	2	1
(l)	What are baghouse filters? Mention one merit and one demerit.	4	2	2
(m)	Mention four benefits of e-waste recycling.	5	2	2
(n)	Write any four effects of e-waste.	5	2	2
(o)	Mention one physical and one chemical water quality parameter.	5	2	3

Part-B**Q2. Answer any two:**

(a)	Distinguish between the primary, secondary, tertiary and quaternary structures of proteins.	1	7	4
(b)	Depict the primary structure of the following peptide codes: FYP, ATF, DCP.	1	7	4
(c)	Classify animal tissues with their functions.	1	7	2

Q3. Answer any two:

(a)	Discuss gene expression: the processes of DNA replication, transcription and translation.	2	7	3
(b)	Discuss competitive, non-competitive and uncompetitive enzyme inhibition.	2	7	3
(c)	Write a short note on the ill effects of drug and alcohol abuse.	2	7	4

Q4. Answer any two:

(a)	What do you understand by sustainability? Discuss the different components of any one ecosystem.	3	7	2
(b)	Discuss the phosphorus cycle in detail.	3	7	4
(c)	Explain decomposition and its steps in an ecosystem.	3	7	3

Q5. Answer the following:

(a)	Explain carbon sequestration and how carbon credits help monetise it as a policy tool.	4	7	4
(b)	Describe the working of a cyclone separator and a baghouse filter used for particulate control.	4	7	3

Q6. Answer the following:

(a)	Write a short note on the circular economy in wastewater treatment.	5	7	4
(b)	Discuss the water quality standards and parameters with reference to drinking water.	5	7	4

Best of Luck

C. V. RAMAN GLOBAL UNIVERSITY, ODISHA
MODEL QUESTION PAPER - SET 5
 END SEMESTER EXAMINATION - BIOLOGY AND ENVIRONMENTAL SCIENCE
 Part B in module order (Q2 = Module 1 ... Q6 = Module 5)

Subject Code: CY202	Subject: BIOLOGY AND ENVIRONMENTAL SCIENCE
Branch: B.Tech - ALL BRANCHES	Time: 3 Hours Full Marks: 100

Answer all Questions from Part - A and Part - B.

Part-A (Answer all questions)				
Q	Part-A (Answer all questions)	CO	Marks	BL
(a)	What is a glycosidic bond?	1	2	1
(b)	Write down any two functions of carbohydrates.	1	2	2
(c)	Differentiate between saturated and unsaturated lipids.	1	2	2
(d)	What are vaccines? Give two examples.	2	2	2
(e)	Write the full form of HIV and AIDS.	2	2	1
(f)	Mention the common uses of any two Indian medicinal plants.	2	2	1
(g)	What do you mean by assimilation in the nitrogen cycle?	3	2	2
(h)	What do you mean by ammonification in the nitrogen cycle?	3	2	1
(i)	Write down four importances of the nitrogen cycle.	3	2	2
(j)	What is carbon sequestration?	4	2	1
(k)	What are the two basic forms of CCS?	4	2	1
(l)	Define noise pollution.	4	2	1
(m)	What is the significance of 3R in waste management?	5	2	2
(n)	What is Integrated Solid Waste Management (ISWM)?	5	2	3
(o)	The carbonate and non-carbonate hardness of a water sample are 300 ppm and 270 ppm respectively. Calculate the amount of permanent hardness.	5	2	3

Part-B**Q2. Answer any two:**

(a)	Describe the structure of the DNA double helix (Watson-Crick model) and differentiate DNA from RNA.	1	7	2
(b)	What are chromosomes and genes? Explain the organisation of DNA into chromatin and chromosomes.	1	7	3
(c)	Elaborate the structure and functions of mitochondria.	1	7	3

Q3. Answer any two:

(a)	Discuss the diseases caused by viruses, bacteria and fungi, and the working principle of antibiotics.	2	7	2
(b)	Discuss immunity, immunization and vaccination.	2	7	2
(c)	Write a short note on COVID-19: its causative agent, spread, symptoms and prevention.	2	7	3

Q4. Answer any two:

(a)	Discuss the ecological pyramids: pyramid of number, biomass and energy.	3	7	3
(b)	What is nutrient cycling? Discuss its importance in an ecosystem.	3	7	3
(c)	Discuss environmental gradients and the tolerance levels of environmental factors.	3	7	2

Q5. Answer the following:

(a)	Discuss the sources and characteristics of noise and the measures for its reduction.	4	7	3
(b)	Discuss the policies and standards for reducing air pollution.	4	7	4

Q6. Answer the following:

(a)	What are the three phases of waste water treatment? Discuss, with diagrams, the working principle of the trickling filter.	5	7	3
(b)	What are biological oxidation ponds? Discuss their working principle with a proper diagram.	5	7	3

Best of Luck

C. V. RAMAN GLOBAL UNIVERSITY, ODISHA
MODEL QUESTION PAPER - SET 6
 END SEMESTER EXAMINATION - BIOLOGY AND ENVIRONMENTAL SCIENCE
 Part B module order shuffled (each module once)

Subject Code: CY202	Subject: BIOLOGY AND ENVIRONMENTAL SCIENCE
Branch: B.Tech - ALL BRANCHES	Time: 3 Hours Full Marks: 100

Answer all Questions from Part - A and Part - B.

Part-A (Answer all questions)				
Q	Part-A (Answer all questions)	CO	Marks	BL
(a)	Write down any two functions of lipids.	1	2	2
(b)	What are glycoprotein and lipoprotein?	1	2	3
(c)	What is a peptide bond?	1	2	1
(d)	Write down any two importances of medicinal plants in modern healthcare.	2	2	3
(e)	What are corticosteroids? Give two examples.	2	2	3
(f)	Cite any two uses of each of Brahmi and Ashwagandha.	2	2	2
(g)	What do you mean by weathering in the Phosphorus cycle?	3	2	1
(h)	What do you mean by an environmental gradient?	3	2	2
(i)	What is a food web? Explain in brief with a schematic diagram.	3	2	1
(j)	What is the normal sound detection range for humans?	4	2	1
(k)	What is the range of man-made and environmental noise?	4	2	3
(l)	Write four harmful effects of noise pollution.	4	2	2
(m)	Calculate the pH of 10^{-3} M HCl and 10^{-3} N HCl solutions.	5	2	3
(n)	What are the three types of water quality parameters? Mention with examples.	5	2	2
(o)	What do you mean by Total Dissolved Solids (TDS)? How can TDS be used to measure the ionic conductance of water?	5	2	2

Part-B**Q2. Answer any two:**

(a)	Discuss the biotic components of the ecosystem and their organisation.	3	7	3
(b)	Discuss in detail the functions of the ecosystem.	3	7	3
(c)	What are trophic levels? Discuss each trophic level with example.	3	7	4

Q3. Answer any two:

(a)	Differentiate between rough endoplasmic reticulum and smooth endoplasmic reticulum, in detail, with a diagram.	1	7	2
(b)	Discuss the classification of carbohydrates into mono-, di-, oligo- and polysaccharides with examples.	1	7	2
(c)	Describe the fundamental characteristics of living organisms.	1	7	4

Q4. Answer any two:

(a)	Describe the major components of a hazardous waste management plan.	5	7	2
(b)	What is e-waste? Cite their common sources and composition. Discuss the steps involved in e-waste recycling.	5	7	2
(c)	Discuss the benefits of e-waste recycling.	5	7	3

Q5. Answer the following:

(a)	What are enzymes? Discuss their structure, characteristics and significance as biocatalysts in biological processes.	2	7	3
(b)	Define fermentation. Discuss the different steps of fermentation.	2	7	2

Q6. Answer the following:

(a)	What is acid rain? Discuss, with reactions, the harmful effects of acid rain.	4	7	2
(b)	What is the greenhouse effect? How do greenhouses work? Give examples of four major greenhouse gases. Discuss the adverse effects of the greenhouse effect.	4	7	4

Best of Luck

C. V. RAMAN GLOBAL UNIVERSITY, ODISHA
MODEL QUESTION PAPER - SET 7
 END SEMESTER EXAMINATION - BIOLOGY AND ENVIRONMENTAL SCIENCE
 Part B module order shuffled (each module once)

Subject Code: CY202	Subject: BIOLOGY AND ENVIRONMENTAL SCIENCE
Branch: B.Tech - ALL BRANCHES	Time: 3 Hours Full Marks: 100

Answer all Questions from Part - A and Part - B.

Part-A (Answer all questions)

Q	Part-A (Answer all questions)	CO	Marks	BL
(a)	What do you mean by structural and functional proteins?	1	2	2
(b)	What is a nucleotide? Name its three components.	1	2	2
(c)	State Chargaff's rule of base pairing.	1	2	2
(d)	Cite any two uses of Neem.	2	2	2
(e)	Name two diseases each caused by viruses, bacteria and fungi.	2	2	1
(f)	Explain the working principle of antibiotics.	2	2	2
(g)	What is an ecological pyramid? Name its types.	3	2	2
(h)	A producer yields a total productivity of 9×10^4 Kcal/m ² /year; it suffers a loss due to internal respiration of 3×10^3 Kcal/m ² /year. Compute the net productivity of the producer.	3	2	3
(i)	What is the difference between environment and ecosystem?	3	2	3
(j)	Determine the sound pressure level if the rms pressure value is given as 30 μ Pa.	4	2	1
(k)	Calculate the sound pressure level in decibels for a given sound pressure of 20×10^3 micro-Pascal.	4	2	1
(l)	What is smog?	4	2	3
(m)	What is the saturation concentration of Dissolved Oxygen (DO) at 20°C and 0°C?	5	2	3
(n)	Mention any two methods of determination of Dissolved Oxygen (DO) in water.	5	2	1
(o)	What is eutrophication?	5	2	2

Part-B**Q2. Answer any two:**

(a)	What is ozone layer depletion? How does CFC affect ozone depletion?	4	7	3
(b)	Discuss the operational principle of an electrostatic precipitator. Comment on its efficiency.	4	7	4
(c)	Write a short note on flue gas desulfurization (FGD).	4	7	3

Q3. Answer any two:

(a)	What is photosynthesis? Discuss the different steps involved in photosynthesis.	2	7	2
(b)	What are biofertilizers? How are they applied? Write down two advantages and two challenges.	2	7	4
(c)	What are pesticides? Write down four advantages and four disadvantages of pesticides.	2	7	2

Q4. Answer any two:

(a)	Elaborate on the different types of organic constituents present in living cells with examples.	1	7	3
(b)	Compare and contrast prokaryotic and eukaryotic cells.	1	7	3
(c)	Describe the functions of the cell membrane, cell wall and cytoplasm.	1	7	2

Q5. Answer the following:

(a)	What is Electromagnetic Radiation (EMR)? Mention its two types. Write the impacts of EMR exposure on human health and explain in brief the mitigation techniques.	5	7	2
(b)	The results from a BOD test, diluted by 100, are given in the table below. Calculate BOD_3 , BOD_5 and BOD_7 .	5	7	4

Q6. Answer the following:

(a)	How does energy flow in an ecosystem? Discuss elaborately.	3	7	3
(b)	Discuss the carbon cycle in detail.	3	7	4

Table for Q5(b):

Time (days)	Diluted sample DO (mg/L)
0	7.95
1	3.75
3	2.75
5	1.80
7	0.75

Best of Luck

C. V. RAMAN GLOBAL UNIVERSITY, ODISHA
MODEL QUESTION PAPER - SET 8
 END SEMESTER EXAMINATION - BIOLOGY AND ENVIRONMENTAL SCIENCE
 Part B module order shuffled (each module once)

Subject Code: CY202	Subject: BIOLOGY AND ENVIRONMENTAL SCIENCE
Branch: B.Tech - ALL BRANCHES	Time: 3 Hours Full Marks: 100

Answer all Questions from Part - A and Part - B.

Part-A (Answer all questions)				
Q	Part-A (Answer all questions)	CO	Marks	BL
(a)	What are enzymes? Give two examples found in the body.	1	2	1
(b)	Define chlorophyll and write two of its functions.	1	2	1
(c)	Write down two functions of carotenoids.	1	2	3
(d)	What is immunity? Differentiate between active and passive immunization.	2	2	1
(e)	What is gene mutation? Give one example of a chromosomal disorder.	2	2	3
(f)	What is a gene, and how does it differ from a chromosome?	2	2	1
(g)	Define ecosystem.	3	2	1
(h)	Mention the types of components of the ecosystem with examples.	3	2	1
(i)	What are the levels of organisation in an ecosystem? Mention the hierarchy.	3	2	2
(j)	Mention any two measures adopted in India to control air pollution.	4	2	2
(k)	Define noise pollution and state the standard permissible noise level in a residential area.	4	2	2
(l)	Compute the amount of carbon dioxide (CO ₂) emitted during the combustion of 16 g of methane (CH ₄).	4	2	3
(m)	What is activated sludge?	5	2	2
(n)	Discuss in brief the working principle of a rotating biological contactor.	5	2	1
(o)	Mention the goals of circular economy.	5	2	2

Part-B**Q2. Answer any two:**

(a)	Write a short note on the circular economy in wastewater treatment.	5	7	3
(b)	Discuss the water quality standards and parameters with reference to drinking water.	5	7	3
(c)	Describe the activated sludge process with a neat diagram.	5	7	3

Q3. Answer any two:

(a)	Discuss Environmental Impact Assessment (EIA).	3	7	3
(b)	What is a food web? Discuss with a suitable diagram.	3	7	3
(c)	What are limiting factors in an environment? Explain a tolerance curve for a single species population existing on a single environmental gradient.	3	7	4

Q4. Answer any two:

(a)	What are vaccines? Discuss the working principle of any three broad categories of vaccines.	2	7	4
(b)	What are steroids? Discuss the functions and significance of steroids with examples.	2	7	4
(c)	Discuss the common Indian medicinal plants and their uses.	2	7	3

Q5. Answer the following:

(a)	Describe the structure and functions of cell organelles such as the endoplasmic reticulum, Golgi apparatus, mitochondria and lysosomes.	1	7	4
(b)	What are the differences between plant cells and animal cells?	1	7	3

Q6. Answer the following:

(a)	Write a short note on noise pollution.	4	7	3
(b)	Describe the main techniques used to control particulate and gaseous air pollutants at industrial facilities.	4	7	4

Best of Luck

C. V. RAMAN GLOBAL UNIVERSITY, ODISHA
MODEL QUESTION PAPER - SET 9
 END SEMESTER EXAMINATION - BIOLOGY AND ENVIRONMENTAL SCIENCE
 Questions fully randomised (structure unchanged)

Subject Code: CY202	Subject: BIOLOGY AND ENVIRONMENTAL SCIENCE
Branch: B.Tech - ALL BRANCHES	Time: 3 Hours Full Marks: 100

Answer all Questions from Part - A and Part - B.

Part-A (Answer all questions)				
Q	Part-A (Answer all questions)	CO	Marks	BL
(a)	What is the difference between environment and ecosystem?	3	2	2
(b)	Write down any two functions of lipids.	1	2	2
(c)	What do you mean by nitrification in the nitrogen cycle?	3	2	3
(d)	What do you mean by Total Dissolved Solids (TDS)? How can TDS be used to measure the ionic conductance of water?	5	2	1
(e)	Name the diseases that appear due to the deficiency of Vitamin A and Vitamin C.	2	2	1
(f)	What is eutrophication?	5	2	2
(g)	What is a glycosidic bond?	1	2	1
(h)	What are the principles of 3R?	5	2	2
(i)	What are ribosomes? Cite their functions.	1	2	3
(j)	What are chromosomes and genes?	1	2	3
(k)	Define a cell and name the scientist credited with its discovery.	1	2	3
(l)	Mention two examples of disaccharides with their constituent monosaccharides.	1	2	1
(m)	What is the significance of 3R in waste management?	5	2	1
(n)	Mention the types of components of the ecosystem with examples.	3	2	3
(o)	What is photochemical smog, and how is it different from smog?	4	2	2

Part-B**Q2. Answer any two:**

(a)	Discuss the water quality standards and parameters with reference to drinking water.	5	7	3
(b)	Discuss competitive, non-competitive and uncompetitive enzyme inhibition.	2	7	3
(c)	Describe the processes involved in the nitrogen cycle.	3	7	2

Q3. Answer any two:

(a)	What are trophic levels? Discuss each trophic level with example.	3	7	3
(b)	Discuss the classification of carbohydrates into mono-, di-, oligo- and polysaccharides with examples.	1	7	3
(c)	Describe the structure of the DNA double helix (Watson-Crick model) and differentiate DNA from RNA.	1	7	2

Q4. Answer any two:

(a)	Discuss in detail the functions of the ecosystem.	3	7	3
(b)	What is Electromagnetic Radiation (EMR)? Mention its two types. Write the impacts of EMR exposure on human health and explain in brief the mitigation techniques.	5	7	3
(c)	Describe the main techniques used to control particulate and gaseous air pollutants at industrial facilities.	4	7	3

Q5. Answer the following:

(a)	What are biofertilizers? How are they applied? Write down two advantages and two challenges.	2	7	3
(b)	Write down the differences between animal cells and bacterial cells.	1	7	3

Q6. Answer the following:

(a)	Explain decomposition and its steps in an ecosystem.	3	7	2
(b)	Describe the functions of the cell membrane, cell wall and cytoplasm.	1	7	2

Best of Luck

C. V. RAMAN GLOBAL UNIVERSITY, ODISHA
MODEL QUESTION PAPER - SET 10
 END SEMESTER EXAMINATION - BIOLOGY AND ENVIRONMENTAL SCIENCE
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Part-A (Answer all questions)				
Q	Part-A (Answer all questions)	CO	Marks	BL
(a)	Explain the working principle of antibiotics.	2	2	3
(b)	Write any four effects of e-waste.	5	2	3
(c)	The carbonate and non-carbonate hardness of a water sample are 300 ppm and 270 ppm respectively. Calculate the amount of permanent hardness.	5	2	2
(d)	Calculate the pH of 10^{-3} M HCl and 10^{-3} N HCl solutions.	5	2	2
(e)	Compute the amount of carbon dioxide (CO_2) emitted during the combustion of 16 g of methane (CH_4).	4	2	2
(f)	What are enzymes? Write down any two applications of enzymes.	2	2	3
(g)	What is immunity? Differentiate between active and passive immunization.	2	2	3
(h)	List any four characteristics of living organisms.	1	2	1
(i)	Write down two differences between rough and smooth endoplasmic reticulum.	1	2	2
(j)	What is a glycosidic bond?	1	2	2
(k)	Write down four important characteristics of lipids.	1	2	3
(l)	State two structural differences between prokaryotic and eukaryotic cells.	1	2	1
(m)	What are biofertilizers?	2	2	3
(n)	Write down any two functions of lipids.	1	2	2
(o)	What are baghouse filters? Mention one merit and one demerit.	4	2	2

Part-B**Q2. Answer any two:**

(a)	Differentiate between rough endoplasmic reticulum and smooth endoplasmic reticulum, in detail, with a diagram.	1	7	3
(b)	Discuss the characteristics, primary source and harmful effects of the criterion air pollutants.	4	7	2
(c)	Describe the major components of a hazardous waste management plan.	5	7	3

Q3. Answer any two:

(a)	Discuss the biotic components of the ecosystem and their organisation.	3	7	4
(b)	What is the greenhouse effect? How do greenhouses work? Give examples of four major greenhouse gases. Discuss the adverse effects of the greenhouse effect.	4	7	3
(c)	Describe the processes involved in the nitrogen cycle.	3	7	2

Q4. Answer any two:

(a)	What is nutrient cycling? Discuss its importance in an ecosystem.	3	7	3
(b)	Define cell. Write down five characteristics of cells and draw a labelled diagram of a cell.	1	7	3
(c)	Describe the main techniques used to control particulate and gaseous air pollutants at industrial facilities.	4	7	2

Q5. Answer the following:

(a)	Describe the functions of the cell membrane, cell wall and cytoplasm.	1	7	2
(b)	Define fermentation. Discuss the different steps of fermentation.	2	7	3

Q6. Answer the following:

(a)	Discuss the ecological pyramids: pyramid of number, biomass and energy.	3	7	2
(b)	What are biofertilizers? How are they applied? Write down two advantages and two challenges.	2	7	2

Best of Luck